COMP 5511: Artificial Intelligence Concepts

Final Project Specifications

Important Notes (Read before you get started!).

1. The project will account for **17%** of the final grade.
2. The project is to be completed individually. Each student needs to develop an agentic AI application or conduct research on fundamental algorithms for AI agents, submit a project proposal, give a presentation, and submit a final report and source code via Canvas.
3. Plagiarism is not allowed. Any suspected cases will be reported to PolyU and processed according to the university's regulations.

Detail Schedule:

23:59 Oct 9, 2026	Project proposal due
23:59 Nov 15, 2026	Presentation slides due
Nov 20 and 27, 2026	Project Presentation in class
23:59 Nov 29, 2026	Final report and code due

Project Overview: AI Agent Applications and Algorithm Research

AI agents are an important step towards applying foundation models to real-world tasks. By combining language understanding and reasoning with planning, memory, and tool use, agents can translate user goals into actions and use environmental feedback to guide subsequent steps. This makes them useful for automating repetitive work and supporting complex tasks across research, education, and everyday digital services [1]. For example, WebAgents illustrate this potential: they observe web pages, plan actions, and interact with websites to carry out tasks such as information gathering and form filling [1]. This project offers two directions: developing an agentic AI application or investigating fundamental algorithms that improve AI agent capabilities. Both directions should include implementation and evaluation with measurable results.

1. Choose either an application project or an algorithm research project. An application project should implement an agentic workflow for a practical task. An algorithm research project may focus on a specific AI agent capability and evaluate the proposed method; a complete end-user application is not required. Algorithm research may address planning, reasoning, memory, tool selection, or multi-agent coordination. Clearly define the application task or research problem and the expected outcome.
2. You may use existing models, APIs, libraries, and frameworks, including the openJiuwen Agent Platform [2], which provides agent development, execution, and debugging capabilities. Platform choice is flexible. Clearly identify your own contribution and acknowledge any external resources used.
3. Demonstrate the implemented application or algorithm and evaluate it using suitable tasks, datasets, or benchmarks.
4. Submit runnable source code with a README explaining setup, dependencies, and how to reproduce the main results.

Final Score for Project

The project accounts for 17% of the final course grade and includes the Project Proposal, Presentation, Final Report, and Implementation.

100% = 10% for proposal + 40% for Presentation + 30% for Final Report + 20% for Implementation.

Project Proposal Submission (23:59 Oct 9, 2026)

Each student should submit a project proposal via the project proposal submission entry on Canvas. Include the project title, student name and ID, and the following content:

Problem and Motivation. Describe the application task or algorithmic research problem and explain its importance for AI agents. For application projects, identify the intended users.

Proposed Approach. Outline the agent workflow or algorithm, relevant models and tools, and your planned contribution.

Implementation and Evaluation Plan. Describe the development or research schedule, data or evaluation tasks, baseline approach, and how you will assess the application or algorithm.

Presentation Slides Submission (23:59 Nov 15, 2026)

Each student should submit presentation slides via Canvas by 23:59 Nov 15, 2026. Minor updates for the presentation are allowed without changing the project title.

Project Presentation (in person, Nov 20 and 27, 2026)

- Each student will do the presentation within 5-7 minutes for the presentation (To be determined). Afterward, there will be a Q&A session --- the teaching team will raise a question in 1-2 min (To be determined).
- The presentation will be arranged in random order by our teaching team. Please let us know if you are not available on Nov 20/27 in advance. Present the implemented application or algorithm and its main experimental results.

Report Requirements (23:59 Nov 29, 2026)

Before we talk about the topic, here are some general requirements:

Report Format. The *project report* is required to follow the format – paper size: A4; font: 12-pt Times New Roman; language: English; margin: 2.5cm (each of the four sides); line spacing: 1.0 (single line).[1] Page limit: 1-page cover page (with only the title and student's information), **at most 10 pages** of contents (*Be careful! It is not the lower bound and we don't set any lower bound for the report length*), and unlimited length of the reference. Table of contents, tables, figures, and appendix sections are not compulsory parts and if included, they should be

[1] Slightly changing the fonts for titles, footnotes, figure and table captions as well as tuning paragraph spacings to allow better display are acceptable.

considered in the limitation of the 8-page content. Also, you are encouraged to use LaTeX for writing.

Deadline. Submission deadline: 23:59 Nov 29, 2026. No late submission is accepted.

Submission Methods. The project report should be submitted via the project report submission entry on Canvas. Please also submit an attached zipped file with the source code used for implementation, experiments, and analyses via the project code submission entry. Include a README with setup instructions, dependencies, required configuration, and example inputs. Please make sure that the code is consistent with the implementation described in the report and is well commented so reviewers can understand the key ideas.

Some tips to get a high score.

1. Using figures and charts for illustration is preferred.
2. For the description of algorithms, using the pseudocode is preferred, but do not attach the real code in the report (Python, Java, C code, etc.).
3. Have a deep analysis of the experimental result. For further deep analysis, you are encouraged to show the results of multiple algorithms on the report, compare the results among them and point out their advantages and disadvantages. If you want to implement and run multiple algorithms, you should introduce the procedure of all the algorithms in detail.

Report Content

We would evaluate the report based on the following content, so the organization of the report is suggested to be this way:

- **Motivation.** Explain why the application task or algorithmic research problem is important.
- **Description.** Describe the agent workflow or algorithm, the models and tools used, and why the approach is suitable for the task or research problem.
- **Implementation.** Explain how you implement the application or algorithm and its key components. Document the key variables and functions in the code.
- **Data.** Present the key statistics and characteristics of the data or evaluation tasks, explain their sources, and justify why they are suitable for evaluation.
- **Experiment Setting, Baselines, and Performance.** Describe the evaluation setup, compare with a suitable baseline, and report measures relevant to the application or algorithm, such as task success rate, accuracy, output quality, execution time, or cost. Include examples or analyses that explain the results.
- **Discussions**. Your thoughts and opinions after analyzing the results e.g., what is the performance of the algorithm, is there any intuitive explanation about why the performance is good/bad, and is there any optimization for addressing this problem. The report should provide some potential insights derived from the project's finding.

Project Topic:

AI Agent Applications and Algorithm Research

You are tasked with either developing an agentic AI application for a practical problem or conducting research on fundamental algorithms for AI agents. Application projects may implement workflows that plan tasks, use tools or resources, and respond to intermediate results.

Algorithm research projects may investigate a specific capability, such as planning, reasoning, memory, tool selection, or multi-agent coordination, without building a complete application. Each student is encouraged to develop an original application, workflow, or algorithmic improvement and explain the design choices. Both directions require a working implementation and an evaluation of effectiveness and limitations.

Choose data or evaluation tasks and baselines that match your application or research problem. Clearly explain how the proposed workflow or algorithm contributes to the results.

References:

Cite the research papers, model and framework documentation, datasets, and other sources used in your project.

[1] Ning, Liangbo, et al. "A Survey of WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models." arXiv:2503.23350, 2025. https://arxiv.org/abs/2503.23350

[2] openJiuwen Agent Platform. Official website. https://openjiuwen.com/en/

About plagiarism

Note: Must avoid plagiarism! Your e-copies will be scanned by **Turn-it-in system!**

* Late submission will be minus 15% per day.

** Turn-it-in score runs very slow near due date. Try to submit 48 hours earlier than the deadline to ensure a possible review of the score result; otherwise you have no chance to modify for resubmission before the deadline.

Green	**0-15%**	Good submission
Yellow	**16% - 35%**	option 1: reduction of 30% marks; option 2: resubmission once within 1 week to attain green zone, -15%
Red	**36% - 100%**	option 1: Zero marks; option 2: resubmission once within 1 week to attain green zone, -15%; yellow zone, -30%